Lecture: Language Modeling

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(Based on slides of Dan Jurafsky)

Probabilistic Language Modeling

- Today's goal: assign a probability to a sentence
  - Machine Translation:
    - P(**high** winds tonight) > P(**large** winds tonight)
  - Spell Correction
    - The office is about fifteen **minuets** from my house
    - P(about fifteen **minutes** from) > P(about fifteen **minuets** from)
  - Speech Recognition
    - P(I saw a van) >> P(eyes awe of an)
  - + Summarization, question-answering, etc., etc.!!

Probabilistic Language Modeling

- Goal: compute the probability of a sentence or sequence of words:

  $P(W) = P(w_1,w_2,w_3,w_4,w_5 \ldots w_n)$

- Related task: probability of an upcoming word:

  $P(w_5|w_1,w_2,w_3,w_4)$

- A model that computes either of these:

  $P(W)$ or $P(w_n|w_1,w_2 \ldots w_{n-1})$ is called a **language model**.

- Better: **the grammar** But **language model** or **LM** is standard

How to compute P(W)?

How to compute P(W)?

- How to compute this joint probability:
  - P(its, water, is, so, transparent, that)
- Intuition: let's rely on the Chain Rule of Probability

- Definition of conditional probabilities

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Rewriting: $P(A \cap B) = P(A)P(B|A)$ or $P(A,B) = P(A)P(B|A)$

- More variables:
  - P(A,B,C,D) = P(A)P(B|A)P(C|A,B)P(D|A,B,C)
- The Chain Rule in General
  - $P(x_1,x_2,x_3,\ldots,x_n) = P(x_1)P(x_2|x_1)P(x_3|x_1,x_2)\ldots P(x_n|x_1,\ldots,x_{n-1})$

If A and B are independent

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A)P(B)}{P(A)} = \mathrm{P(B)}$$

Chain Rule to Compute the Joint Probability

$$P(w_1 w_2 \ldots w_n) = \prod_i P(w_i \mid w_1 w_2 \ldots w_{i-1})$$

P("its water is so transparent") =

P(its) × P(water|its) × P(is|its water) × P(so|its water is) × P(transparent|its water is so)

How to estimate these probabilities?

$$P(\text{the} \mid \text{its water is so transparent that}) = \frac{Count(\text{its water is so transparent that the})}{Count(\text{its water is so transparent that})}$$

Limitations:

- No! Too many possible sentences!
- We'll never see enough data for estimating these

Markov Assumption

- Simplifying assumption:

$$P(\text{the} \mid \text{its water is so transparent that}) \approx P(\text{the} \mid \text{that})$$

$$P(\text{the} \mid \text{its water is so transparent that}) \approx P(\text{the} \mid \text{transparent that})$$

$$P(w_1 w_2 \ldots w_n) \approx \prod_i P(w_i \mid w_{i-k} \ldots w_{i-1})$$

In other words, we approximate each component in the product

$$P(w_i \mid w_1 w_2 \ldots w_{i-1}) \approx P(w_i \mid w_{i-k} \ldots w_{i-1})$$

Unigram Model

- Most simplest model
- Consider a sentence: *“I love programming”*
  - P(“I love programming") = P("I") × P("love") × P("programming")
- Probability of each word is assumed to depend **only on itself and not** on the preceding words
- Hence, the probability of a word appearing in a sequence is **independent** of the words that came before it.

```
fifth, an, of, futures, the, an, incorporated, a,
a, the, inflation, most, dollars, quarter, in, is,
mass

thrift, did, eighty, said, hard, 'm, july, bullish

that, or, limited, the
```

Some automatically generated sentences from a unigram model

Bigram Model

$$P(w_i \mid w_1 w_2 \ldots w_{i-1}) \square P(w_i \mid w_{i-1})$$

Consider the sentence: *"I love programming*"

P(“I love programming”) = P("I") × P("love"|"I") × P("programming"|"love")

Every word in the sequence has a probability that is solely dependent on its preceding word, and not on any other earlier words in the sequence.

N-grams Model

- We can extend to trigrams, 4-grams, 5-grams

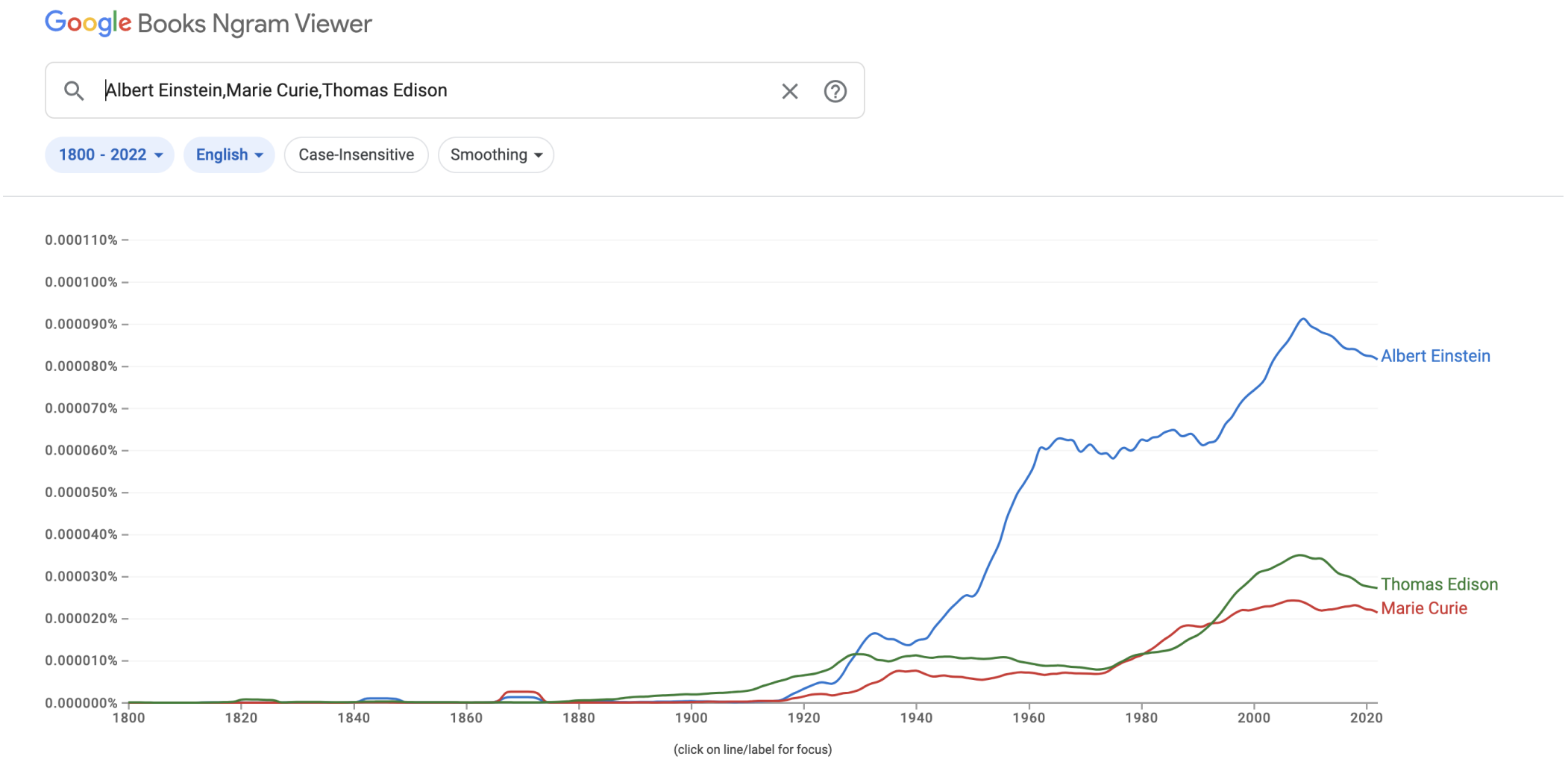
Trigram: $$P(S) = \prod_{i=1}^{n} P(w_i | w_{i-1}, w_{i-2})$$

N-gram: $$P(S) = \prod_{i=1}^{n} P(w_i | w_1, \cdots, w_{i-1})$$

- In general this is an insufficient model of language because language has **long-distance dependencies**:
  - "The computer which I had just put into the machine room on the fifth floor crashed."
  - "The computer which I had just put into the machine room on the fifth **floor** **crashed**."
  - "The **computer** which I had just put into the machine room on the fifth **floor** **crashed**."

- But we can often get away with N-gram models
- Space Complexity: $O(|V|^n)$,
  - where V = vocabulary, n = max. length of sentence
    - E.g.: 475,000 main headwords in Webster's Third New International Dictionary
    - Average English sentence length: 14.3 words
    - A rough estimate: $O(475{,}00014) \sim 3.38 \cdot 1066$ TB
- By applying an N-gram model (say, N =4), we make the model more compact: $O(475{,}000^4)$

N-grams



https://books.google.com/ngrams/

Estimating N-Gram Probabilities

Bi-grams:

The **Maximum Likelihood Estimate**

$$P(w_i \mid w_{i-1}) = \frac{count(w_{i-1}, w_i)}{count(w_{i-1})} \quad \Rightarrow \quad P(w_i \mid w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

Example:

<s> I am Sam </s>

<s> Sam I am </s>

<s> I do not like green eggs and ham </s>

Vocabulary: {I, am, Sam, do, not, like, green, eggs, and, ham}

Maximum Likelihood Estimation

	I	am	Sam	do	not	like	green	eggs	and	ham	<s>	</s>
I	0	2	0	1	0	0	0	0	0	0	0	0
am	0	0	1	0	0	0	0	0	0	0	0	1
Sam	1	0	0	0	0	0	0	0	0	0	0	1
do	0	0	0	0	1	0	0	0	0	0	0	0
not	0	0	0	0	0	1	0	0	0	0	0	0
like	0	0	0	0	0	0	1	0	0	0	0	0
green	0	0	0	0	0	0	0	1	0	0	0	0
eggs	0	0	0	0	0	0	0	0	1	0	0	0
and	0	0	0	0	0	0	0	0	0	1	0	0
ham	0	0	0	0	0	0	0	0	0	0	0	1
<s>	2	0	1	0	0	0	0	0	0	0	0	0
</s>	0	0	0	0	0	0	0	0	0	0	0	0

<s> I am Sam </s>
<s> Sam I am </s>
<s> I do not like green eggs and ham </s>

Vocabulary: {I, am, Sam, do, not, like, green, eggs, and, ham}

Maximum Likelihood Estimation

$P(\texttt{I} \mid \texttt{<s>}) = \frac{2}{3} = .67$ $\quad P(\texttt{Sam} \mid \texttt{<s>}) = \frac{1}{3} = .33$ $\quad P(\texttt{am} \mid \texttt{I}) = \frac{2}{3} = .67$

$P(\texttt{</s>} \mid \texttt{Sam}) = \frac{1}{2} = 0.5$ $\quad P(\texttt{Sam} \mid \texttt{am}) = \frac{1}{2} = .5$ $\quad P(\texttt{do} \mid \texttt{I}) = \frac{1}{3} = .33$

Estimation of the Maximum Likelihood Estimation of the sentence:

<s> I am Sam </s>

P(S) = P (I | <s>)× P (am | I)× P (Sam | am) × P (</s> | Sam)

= 0.67 x 0.67 x 0.5 x 0.5 = 0.112

Maximum Likelihood Estimation

- What if a word occurs that is not part of our vocabulary?
  - “I am John”
- Closed Vocabulary Assumption: The test set can contain only words from our vocabulary.
- Open Vocabulary Assumption: The test set can contain unknown words (out of vocabulary words, OOV) that are not part of our vocabulary.
- In an Open Vocabulary system, unknown words are modeled by adding a pseudo-word <UNK>
  - Convert in the training set any word not in the vocabulary (OOV word) into <UNK>
  - Convert in the test set any unknown word into <UNK>
  - Estimate probabilities for <UNK> like for any regular vocabulary word.

Maximum Likelihood Estimation

<s> I am Sam </s>

<s> Sam I am </s>

<s> I do not like green eggs and ham </s>

Find the probability of the sentence

<s> I am John </s>

P(S) = P (I | <s>) x P (am | I) x P (John | am) x P (</s> | John)

= 0.67 x 0.67 x 0 x 0 = 0

Maximum Likelihood Estimation

1. I love machine learning
2. I love deep learning
3. Deep learning is great
4. Machine learning is fun

Calculate MLE for bi-grams.

Maximum Likelihood Estimation

Example:

- Shakespeare corpus consists of N=884,647 word tokens and a vocabulary of V=29,066 word types.
- Only 30,000 word types occurred (of possible 475,000 referenced by Webster's 3rd New International Dictionary).
  - Words not in the training data ⇒ *P(unknown Word)=0*
- Only **0.04% of all possible 2-grams** occurred.
- The probability of **99.96% of all possible 2-grams is 0**.

Maximum Likelihood Estimation

P(<s> I want english food </s>) =

P(I|<s>) × P(want|I) × P(english|want) × P(food|english) × P(</s>|food)

= .000031

What kind of knowledge?

- P(english|want) = .0011
- P(chinese|want) = .0065
- P(to|want) = .66
- P(eat | to) = .28
- P(food | to) = 0
- P(want | spend) = 0
- P (I | <s>) = .25

Maximum Likelihood Estimation

- We do everything in log space
  - Avoid underflow
  - adding is faster than multiplying

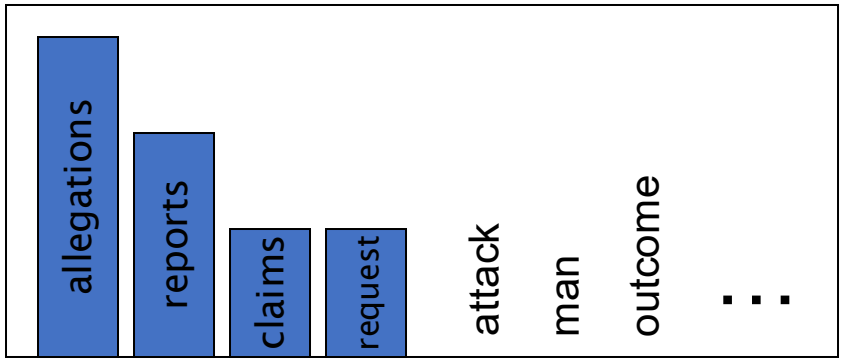
$$\log(p_1 \square p_2 \square p_3 \square p_4) = \log p_1 + \log p_2 + \log p_3 + \log p_4$$

Smoothing

The intuition of smoothing (from Dan Klein)

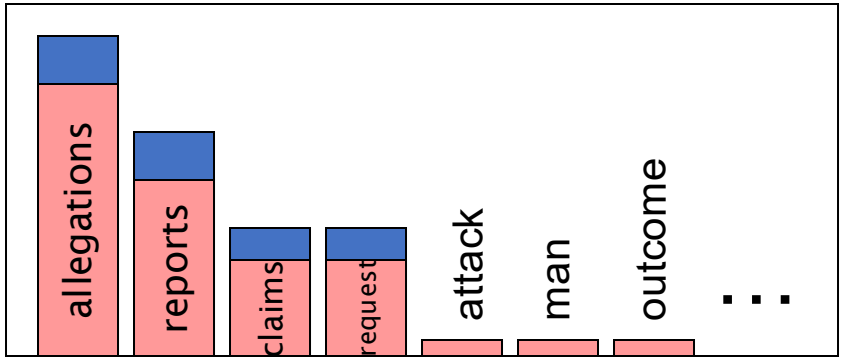
- When we have sparse statistics:

  P(w | denied the)
  3 allegations
  2 reports
  1 claims
  1 request
  7 total



- Steal probability mass to generalize better

  P(w | denied the)
  2.5 allegations
  1.5 reports
  0.5 claims
  0.5 request
  2 other
  7 total



Smoothing

- Also called Laplace smoothing
- Pretend we saw each word one more time than we did
- Just add one to all the counts!

- MLE estimate: $$P_{MLE}(w_i \mid w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

- Add-1 estimate:

$$P_{Add-1}(w_i \mid w_{i-1}) = \frac{c(w_{i-1}, w_i) + 1}{c(w_{i-1}) + V}$$

Berkeley Restaurant Corpus: Laplace smoothed bigram counts

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

	i	want	to	eat	chinese	food	lunch	spend
i	6	828	1	10	1	1	1	3
want	3	1	609	2	7	7	6	2
to	3	1	5	687	3	1	7	212
eat	1	1	3	1	17	3	43	1
chinese	2	1	1	1	1	83	2	1
food	16	1	16	1	2	5	1	1
lunch	3	1	1	1	1	2	1	1
spend	2	1	2	1	1	1	1	1

Laplace Smoothed Bigram Counts

$$P^*(w_n|w_{n-1}) = \frac{C(w_{n-1}w_n)+1}{C(w_{n-1})+V}$$

	i	want	to	eat	chinese	food	lunch	spend
i	0.0015	0.21	0.00025	0.0025	0.00025	0.00025	0.00025	0.00075
want	0.0013	0.00042	0.26	0.00084	0.0029	0.0029	0.0025	0.00084
to	0.00078	0.00026	0.0013	0.18	0.00078	0.00026	0.0018	0.055
eat	0.00046	0.00046	0.0014	0.00046	0.0078	0.0014	0.02	0.00046
chinese	0.0012	0.00062	0.00062	0.00062	0.00062	0.052	0.0012	0.00062
food	0.0063	0.00039	0.0063	0.00039	0.00079	0.002	0.00039	0.00039
lunch	0.0017	0.00056	0.00056	0.00056	0.00056	0.0011	0.00056	0.00056
spend	0.0012	0.00058	0.0012	0.00058	0.00058	0.00058	0.00058	0.00058

Reconstituted Counts

$$c^*(w_{n-1}w_n) = \frac{[C(w_{n-1}w_n)+1] \times C(w_{n-1})}{C(w_{n-1})+V}$$

	i	want	to	eat	chinese	food	lunch	spend
i	3.8	527	0.64	6.4	0.64	0.64	0.64	1.9
want	1.2	0.39	238	0.78	2.7	2.7	2.3	0.78
to	1.9	0.63	3.1	430	1.9	0.63	4.4	133
eat	0.34	0.34	1	0.34	5.8	1	15	0.34
chinese	0.2	0.098	0.098	0.098	0.098	8.2	0.2	0.098
food	6.9	0.43	6.9	0.43	0.86	2.2	0.43	0.43
lunch	0.57	0.19	0.19	0.19	0.19	0.38	0.19	0.19
spend	0.32	0.16	0.32	0.16	0.16	0.16	0.16	0.16

Smoothing in N-grams

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

	i	want	to	eat	chinese	food	lunch	spend
i	3.8	527	0.64	6.4	0.64	0.64	0.64	1.9
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to	1.9	0.63	3.1	430	1.9	0.63	4.4	133
eat	0.34	0.34	1	0.34	5.8	1	15	0.34
chinese	0.2	0.098	0.098	0.098	0.098	8.2	0.2	0.098
food	6.9	0.43	6.9	0.43	0.86	2.2	0.43	0.43
lunch	0.57	0.19	0.19	0.19	0.19	0.38	0.19	0.19
spend	0.32	0.16	0.32	0.16	0.16	0.16	0.16	0.16

- So add-1 isn't used for N-grams:
  - We'll see better methods
- But add-1 is used to smooth other NLP models
  - For text classification
  - In domains where the number of zeros isn't so huge

How to evaluate a Language Model?

Two ways to evaluate: **Extrinsic Evaluation** and **Intrinsic Evaluation**

- **Extrinsic evaluation** captures how useful the model is in a particular task (also known as **in-vivo** evaluation)
  - To compare models A and B
    - Put each model in a real task (such as Machine Translation, speech recognition, etc.)
    - Run the task, get a score for A and for B (such as How many words translated correctly, How many words transcribed correctly etc.)
    - Compare accuracy for A and B
- Limitations:
  - Expensive, time-consuming
  - Doesn't always generalize to other applications

How to evaluate a Language Model?

Intrinsic evaluation captures how well the model captures what it is supposed to capture (also known as in-vitro evaluation): ***Perplexity***

- **Perplexity**
  - Directly measures language model performance at predicting words.
  - Doesn't necessarily correspond with real application performance
  - But gives us a single general metric for language models
  - Useful for large language models (LLMs) as well as n-grams

Perplexity

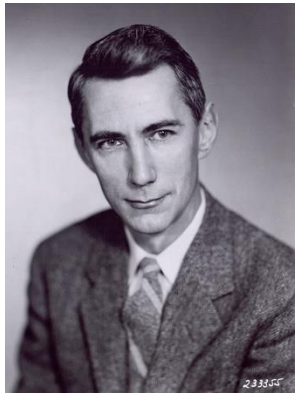
Intuition: A good LM prefers "real" sentences

- Assign higher probability to “real” or “frequently observed” sentences
- Assigns lower probability to “word salad” or “rarely observed” sentences?

The Shannon Game: **How well can we predict the next word**?

- Once upon a ____
- That is a picture of a ____

time	**0.9**
dream	**0.03**
midnight	**0.02**
…	
and	**1e-100**



Claude Shannon

- Unigrams are terrible at this game (Why?)

- A good LM is one that assigns a higher probability to the next word that actually occurs

Picture credit: Historiska bildsamlingen
https://creativecommons.org/licenses/by/2.0/

Perplexity

The best LM assigns high probability to the entire test set

When comparing two LMs, A and B

- We compute P_A(test set) and P_B(test set)
- The better LM will give a higher probability to the test set than the other LM.
- Probability depends on size of test set
  - Probability gets smaller the longer the text
  - Better: a metric that is **per-word**, normalized by length
- **Perplexity is the inverse probability of the test set, normalized by the number of words**

$$PP(W) = P(w_1 w_2 \ldots w_N)^{-\frac{1}{N}}$$

$$= \sqrt[N]{\frac{1}{P(w_1 w_2 \ldots w_N)}}$$

Perplexity

- The inverse comes from the original definition of perplexity from cross-entropy rate in information theory)
- Probability range is [0,1], perplexity range is [1,∞]
- **Minimizing perplexity is the same as maximizing probability**

$$PP(W) = P(w_1 w_2 \ldots w_N)^{-\frac{1}{N}}$$

$$= \sqrt[N]{\frac{1}{P(w_1 w_2 \ldots w_N)}}$$

Chain rule: $$\mathrm{PP}(W) = \sqrt[N]{\prod_{i=1}^{N} \frac{1}{P(w_i | w_1 \ldots w_{i-1})}}$$

Bigrams: $$\mathrm{PP}(W) = \sqrt[N]{\prod_{i=1}^{N} \frac{1}{P(w_i | w_{i-1})}}$$

Perplexity – Weighted Average Branching Factor

- The branching factor of a language is the number of possible next words that can follow any word
- The Perplexity (PP) is referred to as the **weighted average branching factor** and is the most common evaluation metric for N-gram language models

Example: Deterministic language L = {red, blue, green}

Branching factor = 3 (any word can be followed by red, blue, green)

Now assume LM **A** where each word follows any other word with **equal probability ⅓**

Given a test set T = "red red red red blue"

$\text{Perplexity}_A(T) = P_A(\text{red red red red blue})^{-1/5} = ((⅓)^5)^{-1/5} = (⅓)^{-1} = 3$

- But now suppose red was very likely in training set, such that for LM **B**: **P(red) = 0.8 p(green) = 0.1 p(blue) = 0.1**
- We would expect the probability to be higher, and hence the perplexity to be smaller:

$\text{Perplexity}_B(T) = P_B(\text{red red red red blue})^{-1/5} = (0.8 * 0.8 * 0.8 * 0.8 * 0.1)^{-1/5} = 0.04096^{-1/5} = 0.527^{-1} = 1.89$

Perplexity

Wall Street Journal (19,979 word vocabulary)

- Training set: 38 million words
- Test set: 1.5 million words

	Unigram	Bigram	Trigram
Perplexity	962	170	109

Lower perplexity = better language model

Since language model probabilities are very small, multiplying them together often yields to underflow. It is often better to use logarithms instead, so

$$PP(W) = P(w_1 w_2 \ldots w_N)^{-\frac{1}{N}}$$

$$= \sqrt[N]{\frac{1}{P(w_1 w_2 \ldots w_N)}}$$

$$PP(S) = \exp\left(-\frac{1}{n}\sum_{i=1}^{n} \log_e P(w_i | w_1, \ldots, w_{i-1})\right)$$

Evaluation: Accuracy

How to objectively measure the quality of a (classification) experiment?

- Compare your achieved results with a ground truth (gold standard).
- How to achieve a ground truth?
  - Crowd sourcing
- How to compare achieved results with a ground truth?

Scenario:

Binary Classification task: Goal is to predict whether an email is **Spam** (Positive) or **Not Spam** (Negative).

Train set: 10,000 emails, test set: 100 emails and the results are as follows:

Evaluation: Accuracy

Scenario:

Binary Classification task: Goal is to predict whether an email is **Spam** (Positive) or **Not Spam** (Negative).

Train set: 10,000 emails, test set: 100 emails and the results are as follows:

The number of spam emails correctly predicted as spam - 40

The number of not spam emails correctly predicted as not spam - 45

The number of not spam emails incorrectly predicted as spam - 5 Type I error

The number of spam emails incorrectly predicted as not spam - 10 Type II error

	Predicted: Spam	Predicted: Not Spam
Actual/Ground Truth: Spam	40 True Positives (TP)	10 False Negatives (FN)
Actual/Ground Truth: Not Spam	5 False Positives (FP)	45 True Negatives (TN)

Evaluation: Accuracy

	Predicted: Spam	Predicted: Not Spam
Actual/Ground Truth: Spam	40 True Positives (TP)	10 False Negatives (FN)
Actual/Ground Truth: Not Spam	5 False Positives (FP)	45 True Negatives (TN)

Accuracy: Proportion of correctly classified instances out of all instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{40 + 45}{40 + 10 + 5 + 45} \quad = \frac{85}{100} \quad = 0.85$$

Evaluation: Precision and Recall

	Predicted: Spam	Predicted: Not Spam
Actual/Ground Truth: Spam	40 True Positives (TP)	10 False Negatives (FN)
Actual/Ground Truth: Not Spam	5 False Positives (FP)	45 True Negatives (TN)

Precision is the **fraction of retrieved instances that are relevant**

$$precision = \frac{true\,positive}{true\,positive + false\,positive}$$

$$Precision = \frac{40}{45} = \frac{40}{45} = 0.89$$

Recall is the **fraction of relevant instances that are retrieved/predicted**

$$recall = \frac{true\,positive}{true\,positive + false\,negative}$$

$$Recall = \frac{40}{40 + 10} = \frac{40}{50} = 0.80$$

Evaluation: F1 score

	Predicted: Spam	Predicted: Not Spam
Actual/Ground Truth: Spam	40 True Positives (TP)	10 False Negatives (FN)
Actual/Ground Truth: Not Spam	5 False Positives (FP)	45 True Negatives (TN)

F1-score is the harmonic mean of precision and recall, and provides a balance between the two.

$$F_1 = 2 \cdot \frac{precision \cdot recall}{precision + recall}$$

$$= 2 * \frac{0.89 \ * \ 0.80}{0.89 + 0.80} = 0.84$$

Evaluation

	Predicted:Positive	Predicted: Negative
Actual/Ground Truth: Positive	30 True Positives (TP)	20 False Negatives (FN)
Actual/Ground Truth: Negative	10 False Positives (FP)	40 True Negatives (TN)

Compute Accuracy, Precision, Recall and F1 score

Word Meaning

- N-gram methods we've seen so far
  - Words are just strings (or indices w_i in a vocabulary list)
  - That's not very satisfactory!
- Using Logics
  - The meaning of "dog" is DOG; cat is CAT

$$\forall x\ DOG(x) \longrightarrow MAMMAL(x)$$

lemma

mouse (N)

sense

1. any of numerous small rodents...
2. a hand-operated device that controls a cursor...

A sense or "concept" is the meaning component of a word
Lemmas can be polysemous (have multiple senses)

Word Meaning: Synonyms

- Synonyms have the same meaning in some or all contexts.
  - filbert / hazelnut
  - couch / sofa
  - big / large
  - automobile / car
  - water / H_20
- there are probably no examples of perfect synonymy.
  - Even if many aspects of meaning are identical
  - big/large

    my big sister != my large sister

Ask Humans?

word1	word2	similarity
vanish	disappear	9.8
behave	obey	7.3
belief	impression	5.95
muscle	bone	3.65
modest	flexible	0.98
hole	agreement	0.3

SimLex-999 dataset (Hill et al., 2015) gives values on a scale from 0 to 10, from near-synonyms (*vanish*, *disappear*) to pairs that scarcely seem to have anything in common (*hole*, *agreement*)

Word Relatedness

Also called "word association"
Words can be related in any way, perhaps via a semantic frame or field

coffee, tea: **similar**
coffee, cup: **related**, not similar

- Words that
  - cover a particular semantic domain
  - bear structured relations with each other.

hospitals

surgeon, scalpel, nurse, anaesthetic, hospital

restaurants

waiter, menu, plate, food, menu, chef

houses

door, roof, kitchen, family, bed

Computational models of word meaning

Vector semantics

- The standard model in language processing!
- Handles many of our goals!
- Each word = a vector (not just "good" or "w_{45}")
  - Similar words are "**nearby in semantic space**"
  - We build this space automatically by seeing which words are **nearby in text**